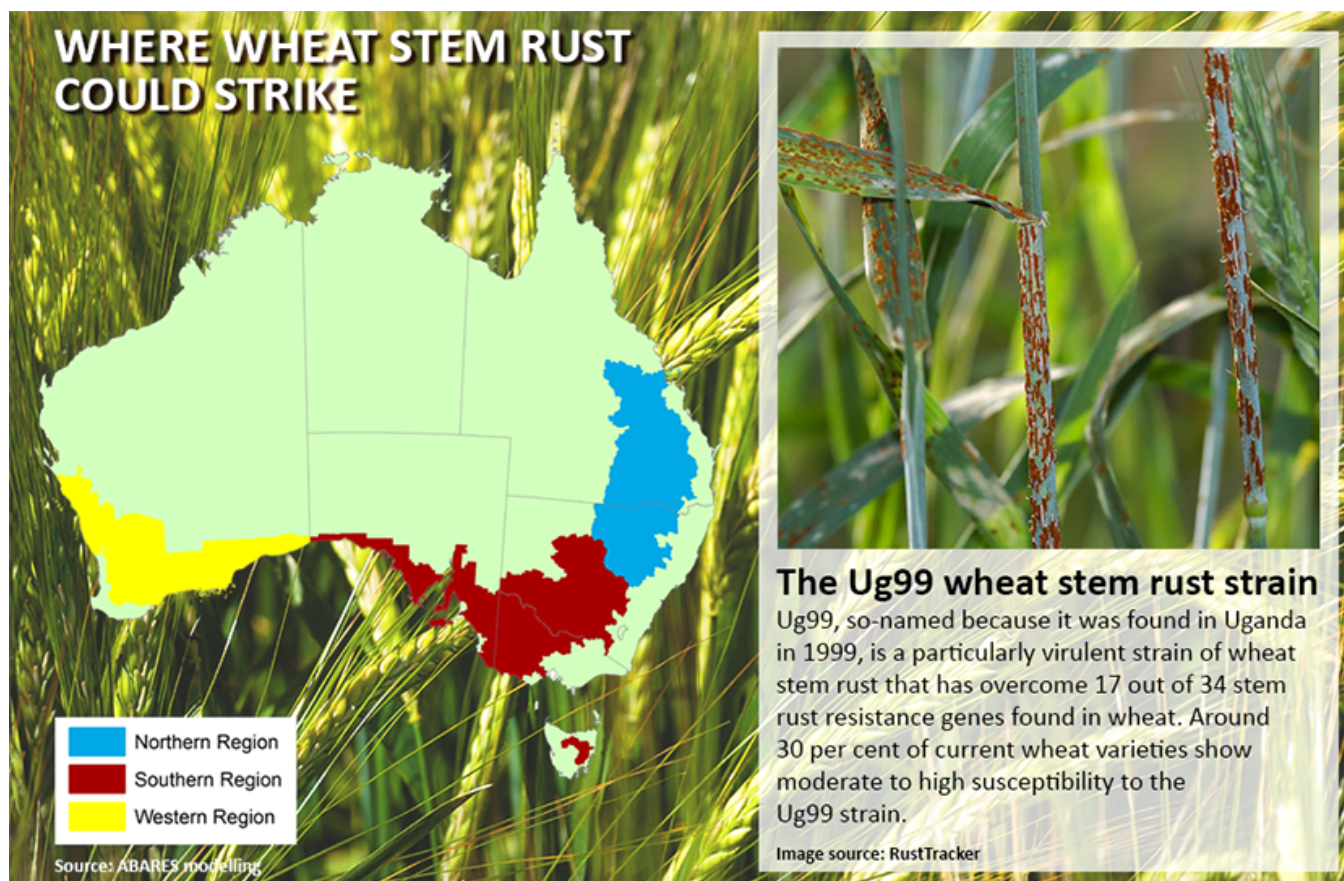




Authors: Donkor Addai, Ahmed Hafi, Lucy Randall, Philip Tennant, Tony Arthur and Jay Gomboso

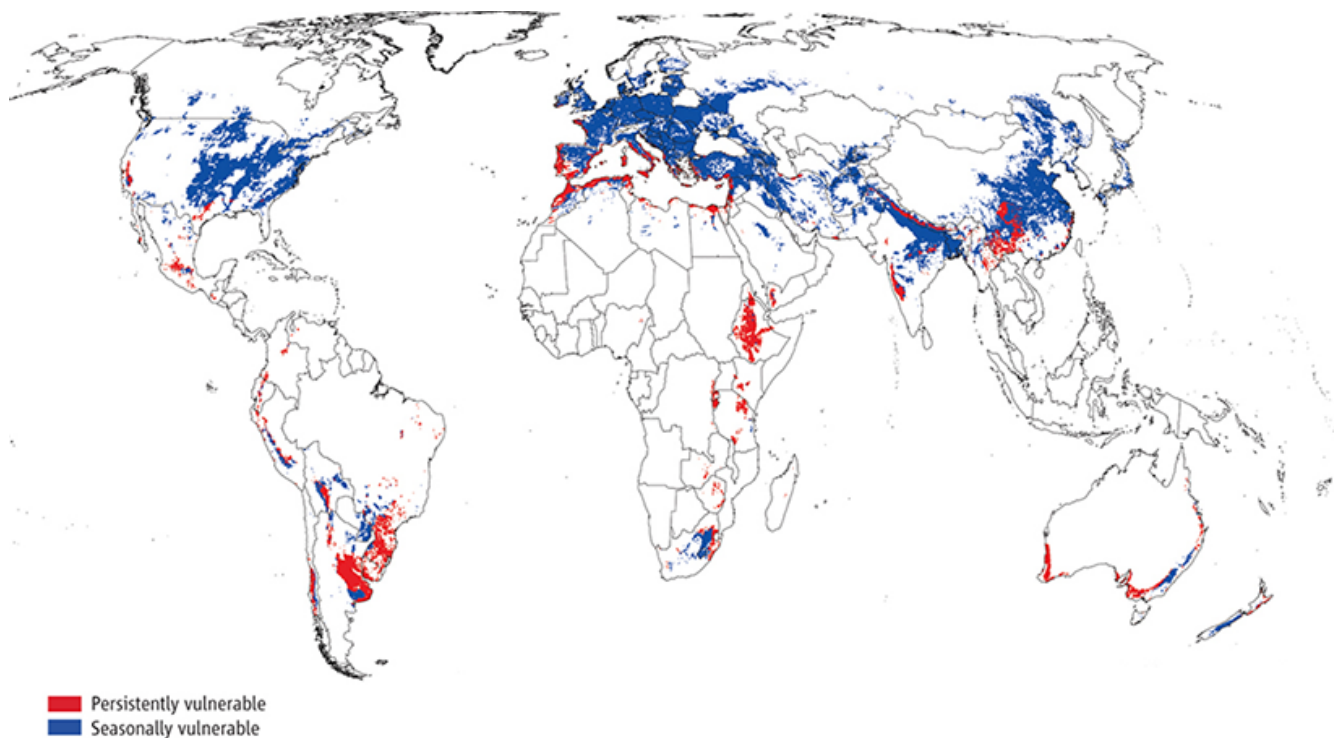
Overview

Wheat stem rust is a fungal disease caused by the *Puccinia graminis* f. sp. *Tritici* (Pgt) fungus that can affect wheat, barley, oat, rye and triticale when seasonal conditions are favourable. The fungus survives on host plants and can spread quickly over large distances by wind, movements of infected plant materials and contaminated farm machinery, equipment and clothing.

Wheat stem rust can attack all above-ground parts of the plant, including the stem, leaves and inflorescence. Infected wheat plants may also produce shrivelled grain. An untreated infection could reduce grain yield by up to 90 per cent.

Wheat stem rust is present in many wheat growing areas throughout the world, and around two-thirds of global wheat growing areas are climatically suitable for the disease. Areas vulnerable to stem rust is shown in Figure 1.

Areas vulnerable to wheat stem rust



Source: Pardey et al. 2013

Wheat Stem Rust has been present in Australia for over a century, with the first documented outbreak occurring in 1889. Since then there have been three subsequent outbreaks in 1899, 1947-48 and 1973. The 1973 outbreak, in south-eastern Australia, is believed to have been the most damaging. Started in South Australia, the outbreak quickly spread to Victoria, New South Wales and parts of Queensland. It reduced the value of the 1973 Australian wheat production by 25 to 35 per cent, and was estimated to have cost the wheat industry between \$1.8 billion and \$2.7 billion (in 2014-15 dollars).

These costs highlight the importance of investment in biosecurity preparedness, containment and eradication measures to minimise the likelihood of future wheat stem rust outbreaks. Such activities include surveillance work (particularly annual surveying), monitoring pathogen populations over time to track potential virulence evolution, and pre-breeding for germplasm resistance.

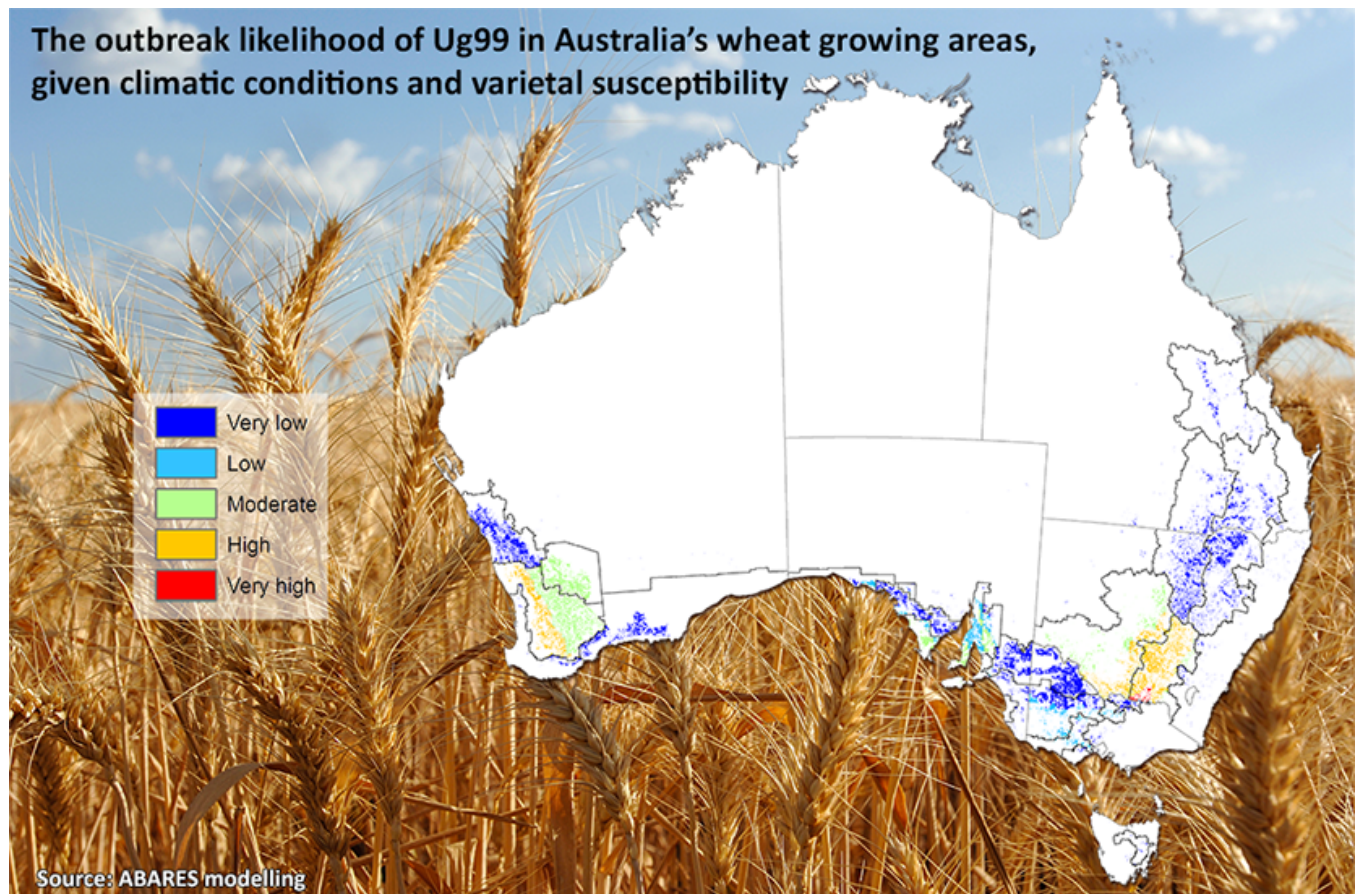
Ug99, so-named because it was found in Uganda in 1999, is a particularly virulent strain of wheat stem rust that has overcome 17 out of 34 stem rust resistance genes found in wheat. Around 30 per cent of current wheat varieties show moderate to high susceptibility to the Ug99 strain. It is not present in Australia, but poses a major risk to the wheat industry, in terms of industry revenue losses and increased production costs, if the strain were to arrive in the country.

This report estimates the economic impact of a wheat stem rust strain Ug99 outbreak on the Australian wheat industry.

Wheat stem rust can first present in Australia via infected plant materials, wind transport over long distances, and contaminated agricultural equipment. The reoccurrence of outbreaks will depend on the suitability of the climate (temperature and moisture) and wheat variety. Evolution of the fungus may produce other virulent strains (both internationally and within Australia) which can also be a source of subsequent outbreaks.

The spread of wheat stem rust depends on the suitability of the climate, susceptibility of wheat varieties, presence of host plants (including those other than wheat), and movement of spores over long distances.

To estimate the likelihood of a Ug99 outbreak for different wheat growing areas, ABARES used their Multi-Criteria Analysis Shell for Spatial Decision Support (MCAS-S)—which uses spatial datasets on climatic suitability ratings and varietal susceptibility ratings as inputs. The outbreak likelihood of each wheat growing area ranged from very low to very high, and is mapped below.



If Ug99 were to establish in a relatively isolated wheat growing region (such as south-west Western Australia), the likelihood of successfully containing and eradicating the fungus would be greater than if it were to establish in areas where wheat growing regions were in close proximity (such as Australia's southern wheat growing regions).

To address uncertainty around extent of spread, ABARES considered three disease spread scenarios of increasing scale:

Scenario 1: an outbreak originating in Esperance, Western Australia, spreading rapidly to all wheat-growing areas in the western region and lasting one year before complete elimination from paddocks. Esperance was chosen because historically, it has been an area prone to wheat stem rust

Scenario 2: an outbreak originating in Ceduna, South Australia, spreading rapidly to all wheat-growing areas in the southern and northern regions of Australia and lasting one year before complete

elimination from paddocks. Ceduna was chosen as it was identified as the starting point of the 1973 outbreak

Scenario 3: the Esperance outbreak, aided by westerly winds, spreads quickly beyond Western Australia to cover all wheat growing areas in the rest of Australia—thereby developing into an Australia-wide outbreak. It also lasts one year before complete elimination from paddocks.

The economic cost of the impact of Ug99 has two components: the cost of the initial outbreak lasting one year and the cost of switching to Ug99 resistant varieties to save losses from future outbreaks of Ug99. When Ug99 spreads uncontrolled following its entry in Australia, the economic costs to the wheat industry may arise from both supply shocks (wheat yield losses and mitigation costs) and demand shocks (wheat import bans by overseas countries). The adoption of Ug99 resistant varieties also has costs: one-off seed cost, annual end-point royalty payments and gross revenue losses from a cut back in production in response to increased cost of production.

The total economic cost, estimated over a 10 year period, increases with the size of the outbreak (Table 1). The costs estimated were \$567 million for the Western Australia outbreak, \$803 million for the south-eastern Australia outbreak and \$1,362 million for the Australia-wide outbreak (Scenarios 1, 2 and 3, respectively in present value terms estimated assuming a 7 per cent discount rate).

Table 1 Economic impact of Ug99 on the Australian wheat industry over 10 years^a—with and without trade restrictions, in 2014–15 dollars

Scenario	Extent of spread	Without trade ban (\$m)	With trade ban (\$m) ^b
1	Western Australia	567	574
2	south-eastern Australia	803	838
3	Australia	1,362	1,403

Note: Economic impacts comprise revenue losses and increased costs of production. a Present value estimated at a 7 per cent discount rate. b based on a hypothetical trade ban from china which last over the outbreak duration.

Source: ABARES modelling

This assessment of the impact wheat stem rust strain Ug99 could have on Australia highlights the importance of our biosecurity system in safeguarding Australia from this deadly plant disease. The study also highlights the importance of maintaining an appropriate level of protection and investment in biosecurity measures aimed at reducing the risk of further wheat stem rust outbreaks in Australia.

[Potential economic impacts of the wheat stem rust strain Ug99 in Australia \(PDF 1.70 MB\)](#)

[Potential economic impacts of the wheat stem rust strain Ug99 in Australia \(DOCX 3.0 MB\)](#)

If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

Source: <https://www.agriculture.gov.au/abares/research-topics/biosecurity/biosecurity-economics/potential-impact-wheat-stem-rust>